# GaN Monolithic Bidirectional Switches

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***Abstract*—Bidirectional switches (BDS) supporting four-quadrant operation serve as the core of various modern converter topologies. These switches are further preferred in monolithically integrated form due to its inherent benefits. In this paper, the benefits of GaN monolithic bidirectional switches (MBDS) over their traditional counterparts, as well as their structure and construction using various approaches, are explored. Following that, key design considerations while using these GaN MBDS, such as substrate termination, proper gate control, and reduction in increase of dynamic ON-state resistance, are studied. Furthermore, the potential benefits of GaN MBDS in advance converter systems are also reviewed.**

***Index Terms*—BDS, GaN, bidirectional switch, HEMT, monolithic integration, reverse blocking, substrate termination, Matrix converter, CSI, T-type inverter, MBDS, gate control**

## I. Introduction

Grid interfaces and motor drives are the two major categories of power electronic converters. Both categories comprise topologies and applications that require bidirectional switches (BDS), which can provide bidirectional current conduction in the ON-state and bidirectional voltage blocking in the OFF-state. Existing power devices lack inherent BDS functionality. To achieve this, we use a combination of discrete power devices. However, conventional BDSs have several limitations related to ON-state resistance ($R_{DS(on)}$ - this abbreviation is used in this paper even for switches with two source terminals), chip area, device count, conduction losses, etc. This led to a shift in the focus towards GaN monolithic bidirectional switches (MBDS), as some of those configurations have $R_{DS(on)}$ comparable to a unidirectional GaN switch while utilizing the same chip area. The monolithic integration also allows for smaller commutation loops, among other advantages. GaN high electron mobility transistor (HEMT) also provides greater flexibility for monolithic integration due to its lateral construction. Furthermore, GaN HEMT, being a wide band gap (WBG) device, has its own advantages of having a high breakdown strength, absence of a body diode, high current density, minimal gate charge, low area-specific $R_{DS(on)}$, high switching frequency operation, etc. [1]. To preserve these advantages of GaN MBDS, certain design decisions need to be made. These include adopting specialized substrate termination techniques like active termination, proper gate biasing schemes, and steps to minimize the dynamic $R_{DS(on)}$ increase due to electron trapping. Modern converters utilizing these well-designed GaN MBDS show significant improvements in their converter efficiencies.

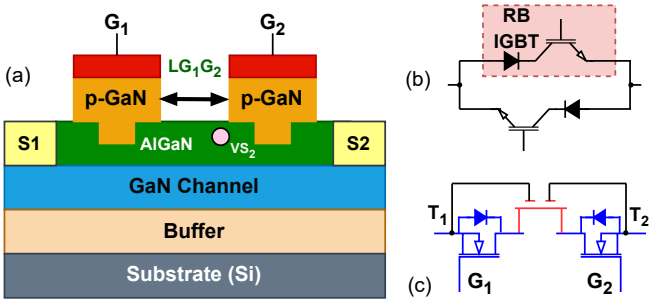



**Fig. 1.** (a) Cross-section of E-mode GaN MBDS (adapted from [7]), (b) RB IGBT based BDS (adapted from [29]), (c) Cascode of D-mode HV GaN MBDS (in red) and LV Si MOSFETs (in blue) (adapted from [27])

## II. Structure of GaN MBDS and its Advantages

Like any other innovative technology, GaN MBDS is developed by overcoming the drawbacks of its predecessors. The evolution and qualitative comparison of various BDSs are described in Table I. Conventional BDSs use 2 IGBTs and 2 diodes. These are cheaper and have a high breakdown voltage ($V_{BR}$), but they comprise multiple devices, resulting in high conduction loss. Antiparallel connection of Reverse-blocking (RB) IGBTs reduces the number of devices to two but still incurs high losses [5] (Fig. 1(b)). Following that, MOSFETs are connected in an inverse series way, leading to further loss reduction, but this still has double the $R_{DS(on)}$ compared to a unidirectional one. Paralleling the second MOSFET reduces $R_{DS(on)}$ but requires four times the chip area. GaN MBDS could be a solution to mitigate all these issues.

GaN based switch topologies come in three varieties: Depletion mode (D mode, normally-ON), Enhancement mode (E mode, normally-OFF), and Cascode mode (D mode with Low voltage (LV) Si FET, normally-OFF). Normally-ON mode is not preferred because of safety concerns, and also demands additional protection circuits. E mode allows for quick turn-on and off switching times, while Cascode mode provides improved thermal performance and can withstand high $V_{BR}$ [16]. GaN HEMTs inherently allow bidirectional current flow but need some modifications for bidirectional voltage blocking [21].

The most popular GaN MBDS is the E-mode dual gate BDS, where a second p-GaN gate referenced to the drain is added in E-mode GaN HEMT. The length of the drift region

**TABLE I.** Evolution and qualitative evaluation of different BDSs (The values mentioned in the cells are based on references in column 2)

| No | Bidirectional switch (BDS) | Component count/Area | Breakdown Voltage($V_{BR}$) | Distinct features |
|---|---|---|---|---|
| 1 | Asymmetric IGBT with diode Bridge | 5/Very High | Very high | Occupy significant space, 3 devices in ON-state path (not preferred) |
| 2 | Asymmetric IGBTs with FW Diodes | 4/Very High | Very high | Still higher losses and IGBT switches suffer from voltage offsets |
| 3 | Antiparalleled RB IGBTs [29] | 2/High | Medium (900 V) | Low leakage current than predecessors and occupy less space but switching losses are very high |
| 4 | Two SiC MOSFETs with Schottky diodes | 4/High | High (1200 V) | SiC MOSFETs comparatively reduce the losses but still component count is high |
| 5 | **GaN Dual gate E mode Monolithic BDS** [7], [17], [23] | 1/Low | Low (650 V) | Nearly the same $R_{DS(on)}$ as unidirectional GaN HEMT utilizing almost the same chip area. Complex gate control due to dual gates |
| 6 | **Antiparalleled RB p-GaN HEMTs (Monolithic)** [20] | 1/Medium | Medium (900 V) | Very low ON-State voltage ($V_{on}$ = 0.63 V) utilizing recessed Schottky drain (34% more efficient than Reverse Conducting type) |
| 7 | **Cascode of dual gate D mode HV GaN MBDS and LV Si MOSFETs (Monolithic)** [27], [18] (Fig. 1(c)) | 1/Medium | High (1200 V achievable) | Floating substrate termination is sufficient leading to high $V_{BR}$, 60% less switching losses compared to SiC counterparts |
| 8 | **pGaN HEMT with recessed anode Schottky barrier diode (SBDs) bridge (Monolithic)** [25], [26] | 1/Medium | High (1100 V) | Recessed SBDs instead of planar, only one gate driver is needed in this BDS, high threshold voltage (1.84 V) |
| 9 | SiC BiDFET (Monolithic)- 2nd Gen [30] | 1/Low | High | Very low $V_{on}$ (0.5 V), Low $R_{DS(on)}$ (27 mΩ) and minimal losses |
| 10 | **Dual GaN BDS with embedded recessed SBDs (Monolithic)** [24] | 1/Low | High (1100 V) | Ultra low $V_{on}$ (nearly 0 V during conduction and 0.4 V during diode mode), very high threshold voltage (2.69 V) |

inside this bidirectional device ($LG_1G_2$ in Fig. 1(a)) is almost the same as that of the unidirectional one, and therefore, $R_{DS(on)}$ is nearly the same. The $R_{DS(on)}$ can be further reduced by the use of a recessed gate structure (etching a gate trench into the AlGaN layer), nearly eliminating the current collapse phenomenon as well [1]. This implies that $R_{DS(on)}$ of GaN MBDS is nearly 50% less than the conventional BDS if parallelization (increase in chip area) is not preferred [1] . Apart from these merits, GaN MBDS has significantly smaller commutation loops, reduced parasitics, and improved PCB integration capability. To evaluate the performance of this dual gate GaN MBDS in terms of switching and conduction losses more accurately, the additional degree of freedom due to the presence of the second gate has to be taken into account [34].

Some shortcomings of this dual gate E mode GaN MBDS, like complex gate control (Section III-B), low $V_{BR}$ (without any modifications), and slightly high ON-state voltage ($V_{on}$), can be overcome with novel GaN configurations (highlighted bold in Table I). Apart from that, the $V_{BR}$ of GaN MBDS can be increased by adopting a vertical GaN structure, increasing the buffer thickness on Si substrate, using multiple field plates of proper dimensions [28], or by utilizing an insulator substrate like Sapphire [22]. GaN MBDS for LV applications are already available on the market (e.g., INN040W048A), but for High voltage (HV), they are only available as prototypes ( [32], [33]) or characterized in TCAD simulations.

## III. Design Guidelines for GaN MBDS Implementation

### A. Substrate termination

The substrate termination technique significantly impacts the ON-state behavior and transient switching of both bidirectional and unidirectional GaN devices. Reliable switching and reduction of degradation, caused by reduced saturation current

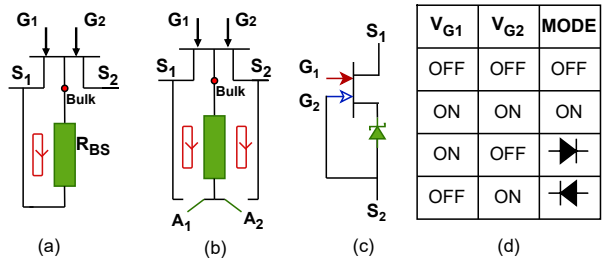



**Fig. 2.** (a) Resistive termination, (b) Active termination ((a) and (b) adapted from [3], [11], [31]), (c) SRB GaN MBDS with asymmetric gates (adapted from [15]), (d) Switching modes of GaN MBDS (adapted from [17])

and increased $R_{DS(on)}$ due to back-gating effects, must be ensured for GaN MBDS to perform competitively [3].

Traditional substrate termination includes passive and floating termination. Passive substrate termination to any source node in GaN MBDS causes asymmetrical ON-state characteristics, while floating termination results in symmetrical but notable $R_{DS(on)}$ increase [3]. Even if nearly zero parasitic inductance is achieved in the gate and power loops by monolithic integration of gate-drivers and proper PCB packaging, passive termination creates an additional parasitic substrate loop (the red loop in Fig.2(a)), leading to unstable switching behavior and issues like "fault turn-on" [11].

*1) Improved substrate termination schemes:*

*a) Resistive substrate termination:* Adding a low-resistive resistor $R_{BS}$ (Fig. 2(a)) in the substrate loop can diminish the instability caused by substrate coupling [11].

*b) Active substrate termination:* Dynamically terminating the bulk node of the GaN MBDS at one of the source nodes, based on the present blocking direction, can be performed using GaN HEMT switches (active) or diodes (semi-floating). Despite the increased circuit complexity, this method can achieve symmetrical bidirectional characteristics with min-

imal increase in dynamic $R_{DS(on)}$. These auxiliary switches (($A_1$), ($A_2$) in Fig. 2(b)) can be monolithically integrated with the MBDS owing to the lateral architecture and shared bulk potential, which further reduces the bulk to the source loop's parasitic and stray inductance [31]. Efficiency measurements of a T-type switching cell with different substrate terminations show the highest efficiencies for active substrate termination, especially in hard switching scenarios [31].

### B. Gate control of GaN MBDS

*a) Gate Biasing:* The switching modes of GaN MBDS based on individual gate biasing is described in Fig. 2(d). To ensure stable ON-state operation, the drain side gate $G_2$ needs to be positively biased by 3 V to 4 V above the total drain potential, while the source side gate $G_1$ can be controlled just like a unidirectional GaN HEMT [1]. This is because higher gate biases ($V_{G2}$>11 V) are required for $G_2$ to achieve similar gate currents as gate $G_1$ with just a 5 V gate bias. The reason for that is that the voltage at the virtual source of the second sub-transistor ($VS_2$ in Fig. 1) increases with higher drain current, which further limits the maximum available current depending on the internal resistive losses [1]. Thus, dual gate GaN MBDS requires different gate biases for each of its gates, making the driver circuitry slightly complex.

*b) Self-reverse blocking control of MBDS:* As discussed, dual-gate MBDS needs two gate control signals, even in fixed current direction topologies like Current source inverter (CSI). To simplify this, a cascode with a LV Si Schottky diode can be used (Fig. 2(c)), where one gate is controlled externally and the other adjusts automatically based on the polarity of the voltage between the source terminals (Self-reverse blocking (SRB)). Asymmetric dual-gate configuration facilitates this realization with normally-OFF characteristic and without any active circuitry [15]. Furthermore, minimizing the parasitics, e.g., by copackaging the diodes with the switches, can eliminate the need for the snubber elements, thereby avoiding their associated side effects.

### C. Reduction of Dynamic $R_{DS(on)}$ increase in GaN MBDS

One of the major advantages of GaN MBDS over conventional BDS is its lower $R_{DS(on)}$ while being compact in size. However, this particular benefit is compromised during high frequency hard switching operations due to the rise in dynamic $R_{DS(on)}$.

*1) Factors responsible for Dynamic $R_{DS(on)}$ increase in GaN MBDS:*

*a) OFF-state blocking voltage:* AlGaN/GaN HEMTs and GaN MBDS are affected by an adverse issue called current collapse (electron entrapment), which causes an increase in dynamic $R_{DS(on)}$ when these devices switch from a high-bias OFF-state to an ON-state [23]. However, a rapid decay of this increased dynamic $R_{DS(on)}$ occurs shortly after turn-on in cases of high OFF-state blocking voltage.

*b) Switching load current:* Increase in load current causes increase in dynamic $R_{DS(on)}$ in both GaN HEMTs and GaN MBDSs [8]. Furthermore, this increase is slightly lower in MBDS compared to a unidirectional one [7].

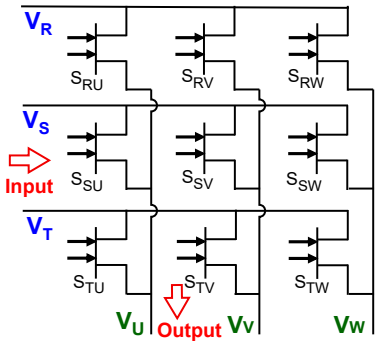



**Fig. 3.** Direct matrix converter employing GaN MBDS (adapted from [2])

*c) Switching frequency:* Even in cases where the total off-time is the same, dynamic $R_{DS(on)}$ increases with an increase in switching frequency. In an experiment conducted on a GaN HEMT [8], the $R_{DS(on)}$ increased significantly when switched from 50 kHz to 400 kHz in continuous conduction mode. This applies to GaN MBDS as well.

*2) Methods to decrease dynamic $R_{DS(on)}$ increase:*

- Using small gate resistors [7].
- Using field plates [14].
- Introducing structural variations: using appropriate buffer layers like SH:FE (Single heterojunction layer with iron buffer) and DH (Double heterojunction) [10].
- Integrating gate driver onto the same chip as MBDS.

## IV. Benefits of GaN MBDS in State-of-the-art Converters

GaN MBDSs are emerging as the top choice for converters that require BDSs. Some state-of-the-art converters [13] that can employ GaN MBDS are

- Third harmonic injected PFC rectifier and Swiss rectifier
- Three-level T-type PFC Rectifier
- Three phase isolated matrix type rectifier
- Three phase T-type three-Level Inverter
- Current source converter (CSC) - Current source rectifier (CSR) + Current source inverter (CSI)
- Indirect and direct three phase AC/AC matrix converter

Direct matrix converter, T-type three-Level Inverter and CSI utilizing GaN MBDSs are reviewed in this paper.

### A. Matrix converters

Matrix converters are among the most efficient AC-AC converters as they provide direct conversion instead of AC-DC-AC conversion as in voltage source converters (VSCs). Also, they do not require a DC link, can provide any frequency to the load, and demand minimum passive components [2].

**Benefits of using GaN MBDS:** The three-phase direct matrix converter requires nine BDSs, which are conventionally built using IGBTs and diodes. The merits of using GaN MBDS in place of this discrete solution are explained briefly in Section II. In addition to those, the GaN MBDS experience very low voltage offsets during the current conduction (minimal offset occurs when operated in diode mode), unlike

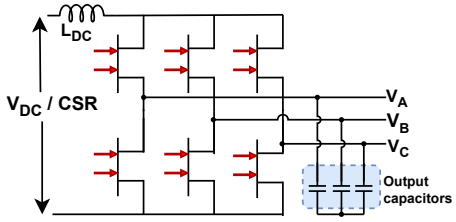



**Fig. 4.** CSI employing GaN MBDS (adapted from [4])

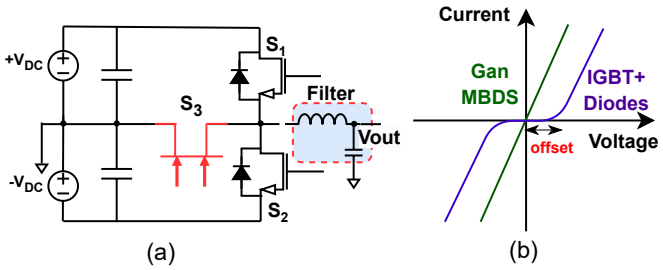



**Fig. 5.** (a) Single phase depiction of 3-level inverter utilising GaN MBDS (adapted from [6], [7]), (b) Offset in IGBT based BDS (adapted from [2])

the huge offsets in IGBT-based ones [2], enabling an efficient AC/AC conversion (Fig. 5(b)). A study on a three phase matrix converter utlizing GaN MBDS (Fig. 3) shows a maximum conversion efficiency of 98% at 1000 W output [2], which is substantially higher than an IGBT-based converter, and the operating losses are nearly halved.

The matrix converter can be further improved in terms of compactness by employing Drive by Microwave Technology (DBM), where all nine BDSs and Radio frequency (RF) rectifiers (to convert RF signals to gate voltages) can be integrated into a single chip [19]. This integration also reduces assembly and wiring costs. This integration is comparatively easier in GaN technology due to its lateral structure. The high heat density issue that usually occurs during the integration of power devices should also not be a problem due to the GaN device´s high temperature tolerance capability [19].

### *B. Current Source Inverter*

Initially, Voltage source inverters (VSIs) have been preferred over CSIs because of their low-cost IGBT switches and lack of demand for reverse-voltage blocking capability, as in CSIs. However, after the usage of WBG devices in the VSI topology, some significant drawbacks are identified, including overvoltages, increased common mode (CM) EMI, high-frequency motor losses, etc. [12], [21]. So, CSIs are taking over today as they inherently have continuous output voltages, double the power density, an integrated LC filter that reduces CM EMI, and a robust DC link inductor instead of the VSI's temperature-limited DC link capacitor [9].

**Benefits of using GaN MBDS:** Conventional CSI uses unidirectional switches with series diodes for bidirectional voltage blocking. When implemented for the kilowatt power range, the large DC link inductor and increased conduction loss due to the series diodes serve as a big disadvantage. These problems can be mitigated by employing GaN MBDS since it has around half the conduction loss compared to switches with series diodes. Due to the reduced conduction loss, the inverter can tolerate a little more switching loss, thereby enabling higher switching frequency operation and reducing the inductor size. A study on a CSC using 60 A, 650V GaN MBDSs (Fig. 4) and a novel synergetic control for a 200 V motor drive (1500 W output) [12] demonstrated 98% efficiency at a 96 kHz switching frequency. Without the synergetic control and GaN MBDS, only a 48 kHz switching operation could achieve the same power and efficiency [12].

Using four quadrant switches instead of unidirectional ones in CSIs also allows for reversing the DC-link current polarity if required, e.g., in distributed DC voltage architectures, eliminating the need for full-bridge front-ends [9].

### *C. T-type 3-level Inverter*

Upgrading from a 2-level to a 3-level inverter (with plus, zero, and minus levels) has been preferred recently, as it significantly reduces the harmonic effects by halving the switching step size. Moreover, the 3-level inverter is more motor-friendly as it creates less stress in motor windings and also minimizes cooling requirements.

**Benefits of using GaN MBDS:** The most critical component in a T-type 3-level inverter is the BDS required to set the neutral point. Even if back-to-back discrete GaN GITs are used as BDS (better than the IGBT-based one), notable oscillations and overvoltage in drain-source voltage ($V_{DS}$) occur due to inherent stray inductance and parasitic capacitance [7]. So, the first step towards reducing these oscillations is the usage of monolithic GaN BDSs (Fig. 5(a)), followed by monolithic integration of the complete single-phase T-type inverter.

A study on a three-level inverter using GaN MBDS exhibits a peak conversion efficiency of 98.7% and a 21% decrease in the conversion loss compared to the IGBT-based one at the output power of 4000 W [6].

## V. Conclusion

Based on the above review, it is pretty evident that GaN MBDS offer significant advantage than conventional BDSs for the converters employing it, especially due to its low $R_{DS(on)}$, compact size, and smaller commutation loops. However, they cannot yet fully replace conventional BDSs, as techniques for characterization of GaN MBDS are not yet matured. Despite proper gate control techniques, dual gate GaN MBDS still has higher gate loop inductance. Therefore, the next step towards enabling better controllability and quicker switching would be to further integrate the gate driver and power stages monolithically onto the same chip as the BDS. Future research can also explore suitable packaging for these GaN MBDS by extracting and comparing the parasitics of different package types employing GaN MBDS.

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